ANKUSH ARORA

Principal Engineer / Technical Lead — Real-Time Systems, XR & Simulation

Zirakpur (Chandigarh Tricity), India | Open to relocation and remote
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Portfolio (30+ projects with video): ankusharoraportfolio.github.io

PROFESSIONAL SUMMARY

Engineering leader with 15+ years building real-time 3D, XR and multiplayer products in Unity and C#. Managed a 25-member cross-functional team delivering VR training and simulation for aviation, healthcare and government clients. Currently architecting a configurable platform that runs around 10 branded marketing games at US trade-show booths from one shared codebase. Known for turning client requirements into clear system designs and shipping them end to end.

KEY ACHIEVEMENTS

- **Gladiator of the Year, Ediiie:** voted best team lead and developer of the year out of 40 team leads and developers.
- Managed a 25-member cross-functional team delivering parallel XR, simulation and game projects.
- Led a VR training simulator with 150+ mechanical engineering modules, such as the jet engine, for an EASA-certified training centre.
- Took Vulcan Tower Defense to 50,000+ downloads within 2 months of launch.

CORE SKILLS

- **Architecture & leadership:** system design, modular and event-driven architecture, state machines, data-driven configuration, team management (25 people), Agile/Scrum and Waterfall delivery, mentoring, client requirement analysis
- **Engine & languages:** Unity (URP, HDRP), C#, Shader Graph, procedural mesh generation, 3D math, performance profiling and optimization
- **XR & simulation:** VR (Meta Quest), mobile AR (LiDAR), WebXR, training and simulation systems
- **Networking & backend:** Photon Fusion, FishNet, Socket.IO, PlayFab, Firebase, REST APIs, Docker, Edgegap
- **Unity tooling:** Cinemachine, Timeline, DOTween, UI Toolkit, Addressables / asset bundles, Unity Cloud Build (CI/CD)
- **AI-Assisted Development:** Claude Code, Model Context Protocol (MCP) / Unity MCP, Prompt Engineering, Generative AI (ChatGPT, Gemini), OpenAI API
- **Platforms:** iOS, Android, Meta Quest, Windows, macOS, WebGL, in-app purchases

PROFESSIONAL EXPERIENCE

Technical Lead — **Fierce Guppy Game Labs** 2025 – Present

US-market interactive games for trade-show and exhibition booths.

- Architected FG Booth Control, a reusable platform that runs around 10 branded marketing games at client booths from one shared codebase.
- Designed a JSON-driven configuration layer so clients change gameplay rules, timings, branding, themes and registration flows without a new build.
- Built the booth flow as a configurable state machine (attract, registration, gameplay, game over, leaderboard) with a companion registration tablet and player queue.
- Implemented local-network host/client discovery so booth screens and registration devices connect automatically on site.
- Integrated PlayFab for leaderboards and player data, with an offline mode and CSV lead export so booths keep capturing leads without internet.
- Built a licence management system to control platform deployments per client.
- Built an AI-assisted production pipeline for 2D games, run in Agile/Scrum sprints: GDD and technical design document first, then detailed concept and asset prompts (ChatGPT), sprite and asset generation (Gemini), development with Claude Code and Unity MCP, automated autopilot play-testing, deployment and marketing-video generation.
- Implemented multiplayer with Photon Fusion dedicated servers (Docker, Edgegap) and telemetry pipelines for analytics and fraud detection.

Lead Unity Developer (Technical Lead, 25-member team) — **Ediiie** 2023 – 2024

XR, simulation and game studio serving international, enterprise and government clients.

- Managed a 25-member cross-functional team of developers, technical artists, QA, designers and a producer across parallel client projects.
- Owned the end-to-end delivery pipeline: requirements, design, art, development, QA and client handover, using Agile/Scrum and Waterfall.
- Led a VR training simulator for an EASA-certified training centre, with 150+ mechanical engineering modules such as the jet engine.
- Delivered NIA Human Anatomy VR for a Rajasthan Government education project, covering all 11 human body systems.
- Led Vulcan Tower Defense (iOS, Android) to 50,000+ downloads within 2 months of launch.
- Delivered Tartarus VR (Meta Quest store), Sinverse multiplayer open-world metaverse, AirportVerse training, server hardware VR training, Jet Engine VR with an OpenAI-powered assistant, and AR apps using LiDAR and WebXR.

Senior Unity Developer — **Orbit Technical Solutions** 2019 – 2023

- Built a fully procedural endless-runner engine that generated tracks, environments and obstacles at runtime, with no hand-made levels.
- Designed Bezier-curve path generation for smooth, endless track layouts.
- Developed runtime mesh generation with custom normals, UVs and texture mapping.
- Created URP visual effects and shaders in Shader Graph, optimized for mobile performance.

Lead Unity Developer — **Techmoweb Experts Pvt. Ltd.** 2018 – 2019

- Led development of a large multiplayer card-game product with Poker and several other card games on one platform.
- Built shared table, lobby and game-rule systems reused across all card games.

Lead Unity Developer — **Igniva Solutions** 2016 – 2018

- Led development of Chuck the Sailor (endless runner) and a multiplayer golf game.
- Mentored developers and owned feature design through release.

Senior Unity Developer — **Dekh! Dekh! India Pvt. Ltd.** 2015

- Built lightweight mini-games embedded inside a chat messaging app.

Senior Associate Unity Developer — **Seasia Infotech Pvt. Ltd.** 2011 – 2015

- Developed 2D and 3D mobile games including endless runners, a jetpack arcade game, Bunny Bazooka, a darts game and car racing.

PUBLISHED WORK

- **Vulcan Tower Defense** — iOS strategy game, 50,000+ downloads in 2 months (App Store)
- **Vulcan Runner** — Android endless runner (Google Play)
- **Tartarus VR** — virtual reality title on the Meta Quest store
- **Sinverse** — multiplayer open-world metaverse (sinverse.com)

EDUCATION

B.Tech, Computer Science and Engineering — Lala Lajpat Rai Institute of Engineering and Technology, Moga (2011)